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Process Engineering

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About

Working on designing, analyzing, and improving large-scale systems with the help of simulations, virtual environments, and data tools.

Top skills: Process Engineering (Chemical & Mining) · Computational Materials (Chemistry & Physics)

Experience



Simulation Module Development Lead

[Aurel Systems Inc.](#)

Jan 2026 - Present

Vancouver, British Columbia, Canada



Mining Engineering Research Assistant

[UBC Norman B. Keevil Institute of Mining Engineering](#)

Jan 2023 - Dec 2025

Vancouver, British Columbia, Canada



Chemical Engineering Research Assistant

[UBC Chemical & Biological Engineering](#)

May 2022 - Dec 2022

Vancouver, British Columbia, Canada



Process Engineering ([Marie Skłodowska-Curie Postdoctoral Fellowship](#))

[Bernal Institute, University of Limerick](#)

Oct 2019 - May 2022

Limerick, County Limerick, Ireland



Research Internship

[Skolkovo Institute of Science and Technology](#)

May 2018 - Oct 2018

Moscow, Moscow City, Russia



Research Assistant

City University of Hong Kong

Nov 2017 - May 2018

Hong Kong, Hong Kong SAR



Visiting Researcher

Institute of Physics & Beijing National Lab for Condensed Matter Physics, Chinese Academy of Sciences

Sep 2017 - Nov 2017

Beijing, China



Summer Internship

Research Institute of Petroleum Industry (RIPI)

Jul 2011 - Aug 2011

Tehran, Tehran Province, Iran

Education



The University of British Columbia

Master's degree, Mining Engineering

Jan 2023 - Dec 2025



University of Tehran

Master's degree, Chemical Engineering (Processes Design)

Sep 2012 - Sep 2014

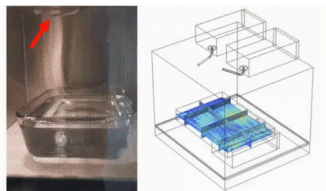


University of Tehran

Bachelor's degree, Chemical Engineering

Sep 2007 - Sep 2011

Projects



Microwave Drying of Minerals: Experimental and Numerical Modeling

Summary

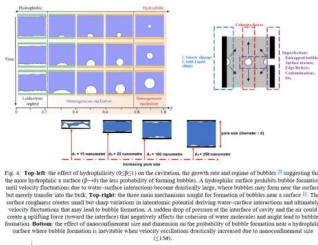
Retrofitting of conventional drying unit operations at a local industrial mining partner. Inspected and evaluated - experimentally and numerically (via Finite Element Modeling in COMSOL) - the feasibility and applicability of microwave-based heating systems at a local mining industrial partner for the retrofitting of conventional drying unit operations.

Tasks Performed

- Conducted experimental and numerical analysis of mineral drying behavior under microwave exposure.
- Utilized finite element modeling (FEM) to simulate heat and mass transfer during drying at various microwave power levels and mineral types.
- Conducted comprehensive energy demand analysis to evaluate potential savings compared to traditional kiln operations.



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Nanocavitation and its Mitigation in Fabricating Artificial Trees

Summary

Investigated nanoscale water flow mechanisms to reduce nanobubble formation and ensure stable, uninterrupted transport in nonporous channels-critical for enhancing efficiency in advanced evaporative cooling systems.

Tasks Performed

- Conducted an in-depth literature review to establish design principles for self-driven water transport at the nanoscale.
- Developed algorithms to quantitatively assess the impact of surface hydrophilicity on nanobubble nucleation and stability.
- Created custom computational routines for high-throughput molecular dynamics (MD) simulations using LAMMPS, tailored for complex fluid-surface interactions.
- Compiled and optimized LAMMPS on UBC's Sockeye HPC environment for scalable, in-house simulation workflows.
- Extended LAMMPS functionality with custom C++ modules to meet unique research needs in nanoscale flow modeling.
- Implemented post-processing scripts in Python for automated extraction and visualization of simulation results, including pressure profiles, density distributions, and flow characteristics.
- Collaborated with a multidisciplinary team to validate simulation results against theoretical models and experimental data, enhancing cross-domain communication and interpretation.



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Fluid Bed Spray Dryer Process Monitoring and Engineering

Summary

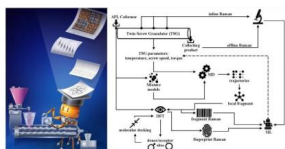
Designed and implemented a data-driven graphical user interface for real-time monitoring and optimization of a fluid bed spray drying process by integrating in-line/offline sensor data streams and advanced analytics into an interactive platform.

Tasks Performed

- Developed an interactive graphical user interface (GUI) in MATLAB for realtime data visualization and diagnostics, supporting both in-line and offline sensor data integration.
- Integrated and processed diverse sensor types including CCD camera feeds (image-based analysis), NIR sensors (unlabeled time-series), Raman spectroscopy probes (localized unstructured signals), and valve states (binary control signals).
- Performed extensive data preprocessing and cleansing to handle high-dimensional and heterogeneous datasets with missing values and sensor noise.
- Applied pattern recognition and signal analysis techniques to identify operational trends, detect anomalies, and support process optimization.

- Designed pipelines for real-time data ingestion and synchronization from multiple sensor sources, ensuring temporal alignment and reliable analytics under dynamic plant conditions.
- Collaborated with process engineers and control specialists to translate sensor insights into actionable process improvements and control strategies.

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Continuous Cocrystalization via Hot Melt Extrusion in Pharmaceuticals

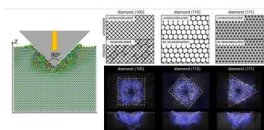
Summary

Developed a data-driven digital twin framework to address low-yield challenges in continuous crystallization, aiming to enhance product quality, optimize production, and reduce waste and operational costs in pharmaceutical manufacturing.

Tasks Performed

- Conducted detailed root-cause analysis of unit operations to identify inefficiencies affecting yield and product purity in continuous crystallization systems.
- Evaluated the influence of critical process parameters-temperature, residence time, screw configuration, and rotation speed-on crystallization outcomes, using both experimental data and simulation insights.
- Designed and refined process strategies to maximize desired product formation, suppress by-product generation, and reduce procurement and disposal costs.
- Built a digital twin using advanced data analytics and implemented a machine learning-based process controller, integrating both real-time (in-line) & historical (offline) sensor data streams - Raman spectroscopy.
- Utilized Density Functional Theory (DFT) and molecular dynamics (MD) simulations to analyze molecular interactions, guiding optimal co-crystal formation pathways and identifying key process descriptors.
- Integrated Raman spectrometer data into a live control system, enabling real-time feedback and control within a continuous manufacturing environment through predictive ML models.

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Machine Learning Inter-atomic Potentials for Materials Discovery

Summary

Aimed to expedite the discovery and characterization of hard materials for use in high-performance environments-such as aerospace, automotive, mining, and manufacturing-by developing and deploying ML-driven interatomic potentials for predictive modeling.

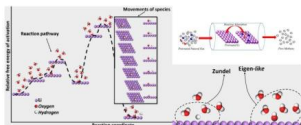
Tasks Performed

- Assessed candidate hard materials for industrial applications, focusing on performance under mechanical stress and durability in extreme conditions.
- Conducted nanoindentation research to evaluate mechanical properties such as hardness and elastic modulus of synthesized materials.
- Developed validation models to discuss experimental results with simulation predictions, extracting insights into material failure modes and defect behavior.
- Implemented and trained Machine Learning Interatomic Potentials (MLIPs) using active learning strategies to improve accuracy with minimal data.
- Automated molecular dynamics (MD) simulations using LAMMPS and density functional theory (DFT) calculations using VASP for large-scale material screening

across multiple HPC clusters.

- Wrote modular and efficient code in Python and Bash, managing environments and version control using Git.

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Design of Adsorptive Systems for Direct Gas Capture and Separation

Summary

Developed a novel process for the direct capture, separation, and solid-state storage of nitrogen and carbon dioxide gases under ambient conditions using moist lithium as a reactive adsorbent, with an emphasis on circular material recovery for sustainable gas handling and sequestration.

Tasks Performed

- Designed and optimized gas capture protocols for ambient-condition adsorption of nitrogen and carbon dioxide on moist lithium enabling safe and efficient conversion into solid-state lithium nitride for storage and transport.
- Applied principles of reaction engineering and separation to evaluate process efficiency, yield, and purity of captured products.
- Conducted Density Functional Theory (DFT) calculations to map reaction pathways between lithium and target gases, identifying favorable thermodynamic and kinetic conditions.
- Developed microkinetic and kinetic Monte Carlo models to simulate reaction dynamics and upscale lab-scale findings for process-scale feasibility.
- Demonstrated on-demand recovery of nitrogen and lithium through electrochemical regeneration, showcasing material circularity and long-term process sustainability.

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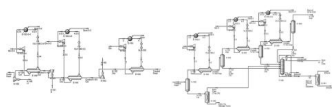
Thermo-kinetic Modeling of Wet Phase Inversion Process for Polymeric Membranes Fabrication

Summary

Developed a comprehensive thermo-kinetic model to simulate the wet phase inversion process for fabricating polymeric membranes, focusing on Multiphysics coupling and accurate prediction of polymeric flat-sheet membrane structure evolution.

Tasks Performed

- Constructed and solved coupled heat, mass, and momentum transport models under non-equilibrium thermodynamics, incorporating moving boundary conditions in multiphase, multicomponent porous systems.
- Formulated and implemented partial and ordinary differential equation solvers (PDE/ODE) to capture the transient dynamics of solvent-nonsolvent exchange and polymer precipitation.
- Wrote custom code in Fortran, MATLAB, and C++ for high-fidelity numerical simulations and sensitivity analyses.
- Validated computational results against experimental measurements, achieving strong agreement in membrane morphology predictions.
- Gained insight into phase separation kinetics, diffusion mechanisms, and the impact of process parameters on membrane performance and structure.



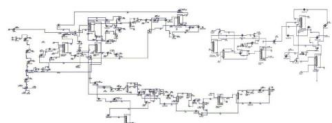
Design of Separation Systems for Ethane and Methane

Summary

Completed a process engineering internship focused on the simulation and optimization of a gas separation unit targeting high-purity ethane and methane recovery, integrating both technical design and economic analysis using industry-standard tools.

Tasks Performed

- Designed, simulated, and optimized a separation process in Aspen HYSYS, achieving efficient separation of ethane and methane under various operating conditions.
- Developed comprehensive Process Flow Diagrams (PFDs) to visualize and document the system configuration, material and energy balances, and process controls.
- Performed equipment sizing and selection for distillation columns, heat exchangers, compressors, and separators based on simulated process data and vendor specifications.
- Conducted cost analysis, including capital expenditure (CAPEX), operating expenditure (OPEX), and utility consumption, to assess economic feasibility and guide procurement strategies.
- Applied exergy analysis and pinch analysis to identify thermodynamic inefficiencies and optimize energy integration, improving process sustainability and reducing waste heat.
- Evaluated and compared equipment procurement options using technical and economic criteria to support decision-making for pilot-scale implementation.



Simulation and Cost Evaluation of Hot Section of BIPC Olefin Plant

Summary

Used Aspen Hysys and Aspen Plus to evaluate retrofitting of industrial scale petroleum refinery complex by producing process flow diagram (PFD), piping/process & instrumentation diagram (P&ID), cost and utility, pinch and exergy.

Tasks Performed

- Simulated existing and proposed process configurations using Aspen HYSYS and Aspen Plus, focusing on optimizing reactor and separation systems for olefin recovery.
- Developed and documented detailed Process Flow Diagrams (PFDs) and Piping & Instrumentation Diagrams (P&IDs) to map unit operations, control loops, and equipment connectivity.
- Performed equipment sizing and specification for heat exchangers, reactors, compressors, and distillation columns based on simulated operating conditions.
- Conducted cost estimation and utility analysis (CAPEX and OPEX) to support retrofitting and procurement decisions.
- Applied pinch analysis and exergy analysis to evaluate and enhance energy integration and thermodynamic efficiency across the system.
- Assessed retrofitting feasibility by integrating performance data, economic viability, and process safety considerations.

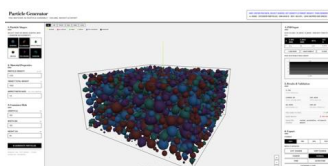
Miscellaneous



Chemical Reaction Systems Analyzer

Summary

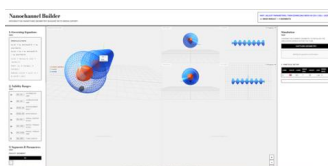
A web-based tool for analyzing chemical reaction systems and Gibbs free energy minimization. This is a demo of the Gibbs Reactor Module in CADSIM Plus (Aurel Systems Inc.). For the simplicity of this demo, all chemicals are treated as ideal gases, and their thermodynamic properties are pulled from the Shomate thermodynamic database provided by the NIST.



Particle Generator and Mesher

Summary

An interactive browser-based tool for particle-filled geometry generation with different shapes and sizes based on the particle size distribution (PSD) and packing characteristics. It can generate 3D granular beds with various shapes and sizes, and volume meshing.



Nanochannel Builder and Mesher (with MD-like Simulations on the Web)

Summary

An interactive browser-based tool for parametric design and 3D visualization of nanochannel geometries, mesh generation in real time, and export. Now supports MD-like simulations.



Convert a Drawing to GPX (Route Pattern Mapper)

Summary

A web app that takes a color-coded pattern image (e.g., a design, logo or similar), overlays it onto a (multi)-city street network (downloaded from OpenStreetMap), extracts GPS coordinates for each color layer, and outputs the result as one or more GPX track files - ready to be used in any GPS-capable navigation apps.



Bike Trainer Dashboard

Summary

Post clavicle fracture fun :) A dynamic, single-page dashboard designed for indoor cycling using Minoura MAG-500 bike trainer.

Tasks Performed

- Gravitational Power: Calculates power required to overcome slope/grade.
- Rolling Resistance: Accounts for tire friction based on total mass.
- Aerodynamic Drag: Estimates wind resistance using rider height and shoulder width.
- Trainer Config: Eight magnetic resistance levels (L1-L8) mapped to specific slopes.
- Drivetrain: Selection for Front Ring (50T/34T) and Rear Cassette (11T-34T) gears.
- Wheel Size: Presets for various tire sizes (700x23C up to 700x32C) with manual circumference input.
- Rider Profile: Defaults are FTP 250W, Mass 85kg, Height 178cm, Shoulder Width

31cm. Recording: Can record a simulated workout and export a GPX file readable by other cycling apps such as Strava.